

MULTI-OBJECTIVE DIET OPTIMIZATION PROBLEM

Using NSGA-II and SPEA2 Evolutionary Algorithms

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1. Introduction

Diet optimization is a complex decision-making problem involving multiple conflicting objectives. Individuals generally expect meal plans that maximize personal preference while minimizing total cost, preparation time, and environmental impact. Since these objectives often conflict with one another, traditional optimization methods are generally insufficient for producing balanced and efficient dietary plans.

In this project, the Multi-Objective Diet Optimization Problem (MODP) was modeled as a Multi-Objective Multidimensional Knapsack Problem (MOMKP). Evolutionary multi-objective optimization algorithms were applied to generate Pareto-optimal meal plans for different types of users. These algorithms are especially suitable for solving complex optimization problems with large search spaces and multiple constraints.

The project uses a MySQL database containing 405 food items together with nutritional information, food groups, user-specific preference scores, and dietary reference intake limits. The implemented system supports both vegetarian and non-vegetarian users and dynamically retrieves all required data directly from the database.

2. Problem Formulation

The objective of the system is to generate a valid daily menu composed of breakfast and lunch+dinner combinations selected from 405 food items. The selected optimization objectives are:

- Maximization of user preference score
- Minimization of meal cost
- Minimization of preparation and cooking time

The mathematical formulation is:

$$f_1(x) = \sum x_i \cdot \text{preference}_i \rightarrow \text{MAX}$$

$$f_2(x) = \sum x_i \cdot \text{cost}_i \rightarrow \text{MIN}$$

$$f_3(x) = \sum x_i \cdot (\text{preparingTime}_i + \text{cookingTime}_i) \rightarrow \text{MIN}$$

Subject to:

$$RLL_j \leq \sum x_i \cdot \text{nutrient}(j, i) \leq RUL_j$$

where:

$$x_i \in \{0, 1\}$$

Here:

- RLL_j represents the lower nutritional bound,
- RUL_j represents the upper nutritional bound,
- and x_j indicates whether a food item is selected or not.

3. Database Structure

The project dynamically retrieves all information from the MySQL database without hardcoding any values. The

database structure contains multiple interconnected tables responsible for foods, nutrients, user preferences, and dietary limits.

4. Chromosome Representation

Each chromosome is represented as a permutation of all 405 food item IDs divided into two independent sections:

[Breakfast Foods | Lunch + Dinner Foods]

- Breakfast section → 94 food items
- Lunch + Dinner section → 311 food items

The two sections are shuffled independently. Genetic operators such as crossover and mutation are also applied separately to each section.

[94 Breakfast Foods | 311 Lunch + Dinner Foods]

5. Decoding Procedure

The chromosome itself does not directly represent the final menu. Instead, a greedy decoding procedure converts genotype information into phenotype solutions.

Breakfast Decoding

The breakfast decoding phase checks only:

- Energy constraints
- Protein constraints

using 35% of daily DRI values.

Lunch + Dinner Decoding

The lunch+dinner decoding phase continues using the accumulated nutrient totals obtained from the breakfast section. During this stage, all five nutritional constraints are checked simultaneously.

Tolerance Values

Tolerance values were used to avoid an empty feasible solution space:

- Upper bound tolerance: +15%
- Lower bound tolerance: -10%

$$RUL_{\text{effective}} = RUL \times 1.15$$

$$RLL_{\text{effective}} = RLL \times 0.90$$

6. Penalty Function

Constraint violations are penalized proportionally in order to reduce infeasible solutions during optimization.

Penalty Equations

$$\text{viol_low}_j = (\max(0, \text{RLL}_j - v_j)) / (\text{RUL}_j - \text{RLL}_j)$$

$$\text{viol_high}_j = (\max(0, v_j - \text{RLL}_j)) / (\text{RUL}_j - \text{RLL}_j)$$

The total penalty value is calculated as:

$$R = 0.7 \times \sum \text{viol_low}_j + 0.3 \times \sum \text{viol_high}_j$$

Final Penalized Fitness

$$R = 0.7 \times \sum \text{viol_low}_j + 0.3 \times \sum \text{viol_high}_j$$

where:

- R represents the total penalty value,
- λ represents the penalty coefficient,
- and v_j represents the nutrient value of the generated menu.

Under-nutrition violations were penalized more heavily because nutritional deficiency is considered more critical than over-nutrition. Therefore, lower bound violations received a weight of 0.7 while upper bound violations received a weight of 0.3.

7. Diversity Mechanism

A valid daily menu should contain foods selected from multiple food groups in order to ensure nutritional balance and menu variety.

In this project, diversity was implemented using a penalty-based mechanism.

Diversity Penalty Formula

$$R_{\text{total}} = R + \alpha \times (1 / (\text{distinct_group_count}))$$

where:

- R represents the original penalty value,
- α represents the diversity penalty coefficient,
- And $\text{distinct_group_count}$ represents the number of different food groups included in the generated menu.

Menus containing fewer distinct food groups receive additional penalties during fitness evaluation. This mechanism encourages the evolutionary algorithms to generate more balanced and diverse meal plans.

The target diversity value was set between 4 and 6 distinct food groups for each daily menu.

8. Evolutionary Algorithms

- Two evolutionary multi-objective optimization algorithms were implemented and compared in this project.

• NSGA-II

NSGA-II (Non-dominated Sorting Genetic Algorithm II) is a Pareto-based evolutionary optimization algorithm. The main mechanisms used in NSGA-II are:

- Non-dominated sorting
- Crowding distance calculation
- Elitism preservation

SPEA2

SPEA2 (Strength Pareto Evolutionary Algorithm 2) is another multi-objective evolutionary optimization algorithm designed to improve convergence quality and solution diversity.

The main mechanisms used in SPEA2 are:

- Strength fitness assignment
- Density estimation
- Archive truncation

SPEA2 uses an external archive structure to preserve high-quality non-dominated solutions throughout the optimization process.

Both algorithms successfully generated Pareto-optimal solutions for the Multi-Objective Diet Optimization Problem.

9. Genetic Operators

The following genetic operators were used during the optimization process:

- **Initialization:** Random permutation
- **Crossover:** OX / PMX
- **Mutation:** Swap mutation
- **Selection:** Binary tournament

Initialization

The initial population was generated using random permutations of food item IDs. Breakfast and lunch+dinner sections were initialized independently in order to increase population diversity.

Crossover

Order Crossover (OX) and Partially Mapped Crossover (PMX) operators were used during reproduction. These crossover methods preserve permutation validity while generating new offspring solutions.

Mutation

Swap mutation was applied by randomly exchanging two genes within the same chromosome section. This operation helps maintain diversity and prevents premature convergence.

Selection

Binary tournament selection was used to choose parent solutions for reproduction. Candidate solutions were compared based on fitness and Pareto dominance criteria.

The crossover probability was set to: $p_c = 0.9$

while the mutation probability was defined as: $p_m = 1/n$

where n represents the chromosome length.

10. Experimental Setup

Experiments were executed separately for:

- User 1 (Non-vegetarian)
- User 2 (Vegetarian)

Experiment Categories

The experiments included:

- User comparison
- Algorithm comparison
- Diversity impact analysis

Experimental Parameters

The optimization parameters used during the experiments are given below:

Parameter	Value
Population Size	50
Generation Count	35
Algorithms	NSGA-II, SPEA2
Crossover Probability	0.9
Mutation Probability	$1/n$

The same parameter settings were used for both algorithms to ensure a fair comparison between the experimental results.

11. Results and Analysis

The generated Pareto fronts clearly demonstrated trade-offs between user preference, meal cost, and preparation time.

NSGA-II generally produced wider Pareto distributions and stronger diversity. SPEA2 demonstrated better convergence in some experiments because of its archive mechanism.

The diversity-enabled runs produced menus with more balanced food group distributions.

Hypervolume metrics were used because the true Pareto front is unknown.

12. Conclusion

This project successfully implemented a Multi-Objective Diet Optimization System using evolutionary algorithms.

The developed system dynamically loads data from a MySQL database, applies custom chromosome representation and greedy decoding, handles nutritional constraints, and generates Pareto-optimal meal plans.

The comparison between NSGA-II and SPEA2 demonstrated that both algorithms are effective for solving MOMKP-based diet optimization problems.

Objective Functions

Objective	Direction
User Preference	MAX
Cost	MIN
Preparation Time	MIN

Nutritional Constraints

Constraint	Nutrient	Unit
C1	Energy	kcal
C2	Protein	g
C3	Carbohydrate	g
C4	Fiber	g
C5	Sodium	mg

Algorithm Comparison

Algorithm	Key Mechanism
NSGA-II	Non-dominated sorting + Crowding distance
SPEA2	Strength fitness + Archive mechanism

References

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